# THIRTIETH ANNUAL UNIVERSITY OF ALABAMA SYSTEM

# APPLIED MATHEMATICS MEETING

## SATURDAY, NOVEMBER 04, 2017

**All sessions will be held on the campus of the University of Alabama at Birmingham in Room 405 of Campbell Hall (Northeast Corner, University Boulevard and 13th Street, Birmingham).**

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| 09:30 | **Refreshments** | Room CH451 |
| 10:15 | **Welcoming Remarks** Department Chair, Rudi Weikard | Room CH405 |
| 10:25 | **Faculty Presentation:** Yanni Zeng (UAB), *Hyperbolic-Parabolic Balance Laws: Structural Conditions and Applications.* | Room CH405 |
| 11:05 | **Graduate Student Presentation:** Tania Hazra (UA), *Stable ADI Scheme with Super-Gaussian Dielectric Distribution and Minimal Molecular Surface.* | Room CH405 |
| 11:25 | **Graduate Student Presentation:** Margaret C. Hock (UAH), *Image segmentation for the analysis of shock images* | Room CH405 |
| 11:45 | **Lunch at area restaurants.** | |
| 1:00 | **Faculty Presentation:** Sivaguru Ravindran (UAH), *Dirichlet Control of Unsteady Navier-Stokes Equations by Boundary Penalty Method* | Room CH405 |
| 1:40 | **Graduate Student Presentation:** Elizabeth Scribner (UAB), *Mathematical Modeling of Brain Tumors Advances Clinical Care, Oncogenesis, and the Use of In Silico Clinical Trials.* | Room CH405 |
| 2:00 | **Faculty Presentation:** Shibin Dai (UA), *Phase-Field Free Energy and Boundary Force for Molecular Solvation.* | Room CH405 |
| 2:40 | **Break/Refreshments** | Room CH451 |
| 3:00 | **Discussions** | |
| | Faculty | Room CH405 |
| | Graduate Students | Room CH458 |

# Abstracts

**Shibin Dai** University of Alabama

**Phase-Field Free Energy and Boundary Force for Molecular Solvation.**

We discuss a phase-filed variational model for the solvation of charged molecules with implicit solvent. The solvation free-energy functional of all phase fields consists of the surface energy, solute excluded volume and solute-solvent van der Waals dispersion energy, and electrostatic free energy. The last part is defined through the electrostatic potential governed by the Poisson-Boltzmann equation in which the dielectric coefficient is defined through a phase field. We prove Gamma- convergence of the phase field free-energy functional to its sharp-interface limit. We also define the dielectric boundary force for any phase field as the negative first variation of the free-energy functional, and prove the convergence of such force to the corresponding sharp-interface limit.

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**Tania Hazra** University of Alabama

**Stable ADI Scheme with Super-Gaussian Dielectric Distribution and Minimal Molecular Surface.**

The work explores the stability impact of the novel unconditionally stable operator splitting methods for solving the time dependent nonlinear Poisson-Boltzmann (NPB) equation for the electrostatic analysis of solvated biomolecules. In a pseudo-transient continuation solution of the NPB equation, the nonlinear term is analytically integrated, so that the difficulties in direct treatment of the strong nonlinearity can be bypassed. We start solving the NPB equation using conditionally stable method, namely Alternating Direction Implicit (ADI) scheme. Our goal is to make the existing ADI schemes stable. Now, for modeling protein electro-statics using implicit methods, we require dielectric properties of the system to be known, in particular, the value of the dielectric constant of protein. In our work, we have applied the Gaussian dielectric constant distribution to ADI scheme (GD-ADI). Also we have used the minimal molecular surface (MMS), that minimizes the surface free energy of the macromolecule in the aquatic environment and it is typically free of geometric singularities. Combination of Gaussian dielectric constant distribution and MMS makes the ADI scheme unconditionally stable. By raising the content of the exponent to a higher power (super-Gaussian dielectric constant distribution) ADI scheme can produce more accurate solvation energy for proteins.

**Margaret C. Hock** University of Alabama in Huntsville

**Image segmentation for the analysis of shock images.**

To compute quantities such as pressure and velocity from laser-induced shock waves propagating through materials, high-speed images are captured and analyzed. Shock images typically display high noise and spatially-varying intensities, causing conventional analysis techniques to have di?culty identi- fying boundaries in the images without making significant assumptions about the data. We present a novel machine learning algorithm that e?ciently seg- ments, or partitions, images with high noise and spatially-varying intensities, and provides error maps that describe a level of uncertainty in the partition- ing. The user trains the algorithm by providing locations of known materials within the image but no assumptions are made on the geometries in the image. The error maps are used to provide lower and upper bounds on quan- tities of interest, such as velocity and pressure, once boundaries have been identified and propagated through equations of state. This algorithm will be demonstrated on images of shock waves with noise and aberrations to quantify properties of the wave as it progresses.

This work was done by National Security Technologies, LLC, under Contract No. DE- AC52-06NA25946 with the U.S. Department of Energy and supported by the Site-Directed Research and Development Program.

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**Sivaguru Ravindran** University of Alabama in Huntsville

bf Dirichlet Control of Unsteady Navier-Stokes Equations by Boundary Penalty Method

In this talk, we wil study optimal Dirichlet boundary control of unsteady Navier-Stokes equation. Dirichlet boundary control, while being practical, is considerably more challenging than other controls in every aspect of control development, from analysis to achieving the control objective. The main difficulty with Dirichlet boundary control is that it is nonvariational and thus it is nontrivial to identify a suitable function space framework without using appropriate boundary lifting. The author presents a boundary penalty approach to treat the Dirichlet control of unsteady Navier-Stokes equation. The boundary penalty approach studied in the paper allows one to work with L2-controls. Unlike other L2-control space approaches, this does not lead to optimality conditions that involve a normal derivative of the adjoint variable on the boundary. Existence of optimal control for a class of optimal control problems is established. The asymptotic behavior of solution, with respect to the penalty parameter, is studied. In particular, we prove convergence of solutions of penalized control problem to the corresponding solutions of the Dirichlet control problem, as the penalty parameter goes to zero. We also derive an optimality system and determine optimal

solutions. Feasibility and applicability of the approach are illustrated by numerically solving a canonical flow control problem.

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**Elizabeth Scribner** University of Alabama at Birmingham

**Mathematical Modeling of Brain Tumors Advances Clinical Care, Oncogenesis, and the Use of In Silico Clinical Trials.**

Glioblastoma multiforme (GBM) is a uniformly fatal form of brain cancer that can exhibit unpredictable responses to different rate-reducing therapies. Here, we construct a mathematical model of brain tumor growth and utilize in silico clinical trials to investigate the unusual behavior of GBM. We also use the model to study the progression of grade 2 and grade 3 tumors to higher grades like GBM. Simulations lead to the identification of a previously unknown pattern of progression under anti- angiogenic (AA) therapy, which we call ?Expanding FLAIR + Necrosis.? Model analysis predicts that clinical features of this growth would include expanding regions of brain necrosis and low-concentration invasive tumor cells, which we find in the MRIs of 11 out of 70 patients treated with AA. This new patient phenotype suggests key hypotheses on the pathogenesis of tumor progression. Namely, we identify tumor motility as an important biomarker because it appears to exert a significant impact on patient prognosis, tumor progression, the efficacy of different therapies, and overall survival times. In studying the development of tumor grades, we also develop a new concept in cancer genetics?the idea of natural tumor evolution vs. transformation. Both model simulations and clinical data support the conclusion that the natural evolution from grade 3 glioma to GBM could occur in a period of a few months. We also identify a subgroup of patients whose tendency to recur at the same grade is associated with specific molecular mutations. We hypothesize that progression of these gliomas to higher grades necessitates the accrual of molecular changes affecting key rates, which we call a vertical transformation. All of these findings serve to advance the basic medical sciences, particularly the field of oncogenesis. They also herald the future use of in silico clinical trials in drug discovery by showing how computer models can safely and effectively evaluate therapeutic choices. Finally, our research promotes a multidisciplinary approach to cancer research by demonstrating the capability of mathematical modeling in testing hypotheses, validating phenotypes, and predicting clinical data.

**Yanni Zeng** University of Alabama at Birmingham

**Hyperbolic-Parabolic Balance Laws: Structural Conditions and Applications.**

We study a general system of hyperbolic-parabolic balance laws in $m$ space dimensions ($m \geq 1$). The system has physical viscosity matrices. Besides, there is a lower order term to account for relaxation, damping or chemical reaction. The viscosity matrices and the Jacobian matrix of the lower order term are rank deficient. We propose a set of structural conditions, which include Kawashima-Shizuta condition. We study the Cauchy problem around a constant equilibrium state. The conditions allow us to establish the existence of solution global in time and obtain optimal $L^p$ asymptotic decay rates ($p \geq 2$) of the solution to the constant equilibrium state. In the case of multi space dimensions, we can further study time-asymptotic behavior and obtain optimal $L^p$ decay rates to the asymptotic solution. These general results recover the existing results in literature on hyperbolic-parabolic conservation laws and hyperbolic balance laws, respectively, as two special cases. The proposed structural conditions are sufficiently general for applications to physical models such as physical gas flows. In particular, they apply to gas flows with translational and vibrational non-equilibrium.